PSYCON: CONTEXTUAL COMMUNICATION BEHAVIOR AND EVIDENCE-LINKED PSYCHOLOGIST OBSERVATIONS

SEPARATE TWO-PAGE RESEARCH SYNOPSIS

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ABSTRACT

Group-discussion behavior depends on what happened immediately before, whether a participant had a fair opportunity to respond, and the person's earlier behavior. A brief contribution or an overlapping turn alone cannot establish anxiety, disengagement, or personality. PSYCON investigates whether speaker-specific conversational measurements can predict contextual observations reviewed by psychologists, and whether earlier personal history improves evidence-supported communication reports. Its local research instrument processes audio, produces anonymous speaker intervals and transcripts, retains timestamped exchanges, and connects reviewed participant mappings to a versioned A–T marksheet. Ratings distinguish 0–4 observations, insufficient opportunity, and missing answers. The planned behavioral evaluation measures independent rater agreement, held-out per-item prediction error, and blinded report grounding and usefulness. Participant/source grouping prevents related recordings from crossing evaluation boundaries. The implemented report comparison uses transcript-only, structured-context, and full-PSYCON inputs with matched local model settings. Human behavioral outcomes remain pending. Retained engineering evidence covers eleven recordings and 85 person-recording slots: guarded PSYCON yielded 57 ready profiles and guarded Nemotron yielded 51 in each tested configuration. These describe coverage, not identity accuracy or psychological validity. The 76 imported version 5.0 forms contain 750 numeric, 1,662 N/O, and twenty missing answers. The source-score pilot uses 423 ratings from forty examples across nine recordings and five known groups. Compact grouped development achieved 62.19% agreement; new nested pooling achieved 65.67%, exactly matching the training-mode predictions while reducing score-3 recall. No candidate was promoted, and independent validation remains unavailable. The contribution is an auditable behavioral measurement workflow. The next study must establish permission, reproducible human scoring, and independent evaluation before claims of predictive or communication benefit.

1 Introduction and behavioral question

The question is whether measurements and context predict reviewed A–T observations on held-out people/sources, and whether earlier history improves supported reports. The rubric concerns discussion tracking, contribution structure, turn-sharing, challenge, and delivery change. These are operational areas, not diagnostic factors. Reduced speech after challenge needs the event, earlier behavior, later response, and opportunity. Source context and measurement validity matter because a plausible model description cannot supply its own human validation.

2 Innovation and objectives

The contribution links participant, source time, exchange, reviewer, and revision in an auditable measurement workflow. H1 predicts lower held-out item error than the training-median baseline. H2 predicts better blinded grounding/usefulness without increased overclaiming. A history-only experiment must isolate history from the evidence/references also added by full PSYCON.

3 Methodology and controls

The unit is a participant within a discussion. A reviewer confirms person/participant/speaker mapping. Nineteen author-supplied cross-recording links reduce 85 slots to at most 66 people and impose six provisional recording groups; further overlaps remain unknown. Linked recordings must share a partition. Recordings retain context, time, permission, and hashes. Audio processing conservatively attributes speech and preserves evidence; timing/marker proxies do not prove intent.

Psychologists supply scores, opportunity, confidence, recording support, and evidence, followed by independent review. Disagreement remains available before adjudication. Positive scores need timestamps; event-linked items need event/response windows, and Q–T need an earlier same-session baseline. Item S lacks predictors; visual-only T cannot train the audio model. Version 5.0 imports remain separate from v4.0; transcription and validity review are pending.

Frozen datasets group repeated people and related sources before splitting. Training-only preprocessing precedes per-item logistic classification; validation selects regularization, with test data separate. Fitting minimums are engineering gates, not a power calculation. Nine private signed sheets document voluntary participation. The authors confirm recording/training permissions, applicable parental permission and prior approval in the complete records; these details and recording linkage have not been independently checked. Analysis needs item error against the training-median baseline, weighted agreement, exclusions, abstentions, and grouped uncertainty. Reports need blinded ratings, randomized order, and a primary outcome. Language, topic, moderation, opportunity, and recording quality are confounds.

4 Current results, limits, and next study

The audiovisual comparison retained 57 PSYCON profiles versus 51 per Nemotron mode out of 85 slots, not unique people. The source-score pilot uses forty examples, nine recordings, five known groups and 423 ratings across G, I and P1–P12. Training agreement of 97.9% is resubstitution. Original grouped development matched 230/402 ratings (57.21%, MAE 0.562) with twenty-one abstentions. Compact selection reached 250/402 (62.19%, 0.490), versus training median 259/402 (64.43%, 0.460) and mode 264/402 (65.67%, 0.453). The previous agreement-selection experiment reached 247/402 (61.44%, 0.498). A new bounded inner-group-selected pooled procedure exactly matched mode predictions, reducing score-3 recall from 18.45% to 4.85%. A fixed 67.16% diagnostic predicted score 4 on 397 ratings and was not selected from outer results. Compact remains the research reference; no candidate was promoted. All historical predictions refitted exactly. Other procedures score all 423 with no abstentions: compact 61.23%, median 63.36%, mode/new 64.54%. Score 0 is unobserved; all-rating counts for 1–4 are 24,24,109,266. Twenty-one unresolved participant-session identities and absent independent review, opportunity, confidence and structured evidence prevent independent generalisation or a numerical ceiling. No final test supports 85%. The report example has no eligible history or blinded usefulness result, so H2 remains open.

Next work must verify transcriptions, permission/mapping, and independent rater reliability, then freeze an independent sample. A history-only ablation should hold transcript, evidence, references, model, and decoding fixed. Poor agreement, no baseline improvement, or increased unsupported claims would challenge the hypotheses. Hardware/3D/KiCad work is paused future measurement research. Voluntary communication review is a potential use; diagnosis, disciplinary scoring, causation, and worldwide benefit are unsupported.

5 Acknowledgements and selected references

School psychologists supplied observation reference material; open-source authors supplied speech tools. Private identities and school/location identifiers are omitted. AI assisted this working draft; authors must independently rewrite applicable submission text and verify claims, references, contributions, consent, and approvals. No poster is included.

Sacks, Schegloff and Jefferson (1974), turn-taking, doi:10.2307/412243. Cronbach and Meehl (1955), construct validity, doi:10.1037/h0040957. Cohen (1968), weighted agreement, doi:10.1037/h0026256. Full references and engineering sources accompany the separate paper. Study framing follows the ISEF science judging criteria; human-study requirements follow the Human Participants rules.